

David Quist

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Education

McMaster University – B. Eng in Mechanical Engineering

Expected June 2030

Toronto Metropolitan University – BS in Mechanical Engineering (4.1/4.33 GPA)

Sept 2025 - April 2026

- GPA: 4.1/4.33
- Activities: Toronto Met Racing Member, 2x Dean's List Recipient

Skills

CAD: SolidWorks (CSWA), Fusion360

Programming: Java, C, C++, Javascript, HTML, CSS, MATLAB

Additional: Microsoft Office, OFSAA Swimmer, SFA, National Lifeguard, Instructor's

Projects

6-DOF Robotic Arm

github.com/quistty/6dofarm

- Designed 6-DOF robotic arm with 500 mm reach, 2 kg continuous payload, ± 2 mm repeatability
- Engineered 3 reduction systems using multi-stage planetary gearboxes and pulley reductions reducing up to 84:1.
- Developed off-axis absolute magnetic encoder system using alternating-polarity magnets to provide $< 1^\circ$ angular resolution at 500 Hz sampling for closed-loop position verification, calibration, and future ROS 2 motion planning.

Skyblock API Scraper

github.com/quistty/rub

- Developed Javascript/Node.js Discord Bot that retrieved current Hypixel Skyblock Bazaar pricing data in 30 ms ping to calculate real-time profit margin for tradeable items
- Implemented command-based user interactions and API-driven price calculations, deploying and maintaining the open-source application through documented installation and configuration workflows

Experience

Lifeguard/Swim Instructor/Examiner, City of Richmond Hill – Richmond Hill, ON

Feb 2024 – Sept 2026

- Instructed hundreds of swimmers across varying ages and skill levels, adapting lessons to individual needs and helping students progress toward higher swim certifications.
- Prepared and examined swimmers for Bronze Medallion and Bronze Cross standards, providing targeted feedback that supported successful certification and improved swimming technique, confidence, and safety awareness.

Lifeguard, PPL Aquatics – Toronto, ON

Aug 2025 – Aug 2026

- Oversaw swimmers and pool conditions at government housing facilities, responding to emergencies and providing first aid as needed to maintain a safe aquatic environment.